

telecom, solar and LED continue to be demand-generating areas and the core growth continues from the industrial automation and instrumentation segments.

Further, advances in technology such as high-density interconnection (HDI) technology would enable more interconnection functions per unit area, which has the potential of having a high growth impact on the market for advanced-level applications. For example, according to Muniswamy, availability of the Internet to remote areas, changing lifestyles with increased purchasing power of the middle class and rapid growth of the e-commerce industry is expected to generate demand for PCBs required for mobile and portable devices, cloud computing servers and networking devices.

"Mobile computing, cloud computing and the Internet of Things (IoT) are the latest fads in the electronics industry, and we are seeing the emergence of local brands in mobile computing, which includes smartphones, tablets and laptops, among others. Policy changes are forcing local brands to start manufacturing in India, and this should help the local PCB manufacturing ecosystem and create millions of jobs," he adds.

"In recent years, consumer demand and regulatory concern for low carbon emission, energy saving and environmental protection has attracted greater attention towards LED lighting. LED manufacturing segment in the country is poised to grow at a very high rate. This, in turn, has the potential to open up a huge demand potential for PCBs," says Rajoo Goel, secretary general, ELCINA.

According to Anuraag Dhoot, managing director, Epitome Components, presently, the best bet for Indian PCB manufacturers is in the area of supplies to energy meters, UPS systems and automotive electronics. These are the growing areas that have good potential for growth and expansion for PCB manufacturers.

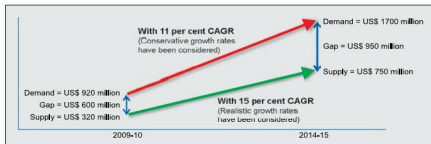


Fig. 1: Demand-supply gap in Indian PCB industry (Source: IPCA)

Steps to promote indigenous ecosystem for PCBs

- ▶ Imposing high import tariff; nil duty on raw material imports
- ▶ Interest subsidies/low interest loans; quicker reimbursement of CENVAT
- ▶ Provision of export subsidies
- ▶ Elimination of inverted duty structure
- ▶ Imposing anti-dump duty/safeguard duty

Source: An IESA - Frost & Sullivan report

"Given that the automotive industry in India is growing steadily and every car needs two to 15 PCBs depending on features and functionality requirements, demand for PCBs from this segment is bound to grow," opines Paresh Vasani, managing director, Circuit Systems India Ltd.

In addition, the recent thrust on indigenisation of strategic electronics manufacturing will likely fuel the demand for high-grade multi-layer PCBs.

Demand-supply gap

Currently, most domestic PCB consumption in the country is served through imports. This does not bode well for the economic health of the industry in the long term. "Currently, almost 80 per cent to 85 per cent of the country's PCB requirements are met through imports. To boost electronics manufacturing in the country, it is important to have adequate PCB manufacturing capacities that meet the need of the industry at competitive prices, thereby preventing excessive dependence on imports," says Goel.

According to an IPCA report, there is a gap of around US\$ 950 million (Fig. 1) between the demand and supply of PCBs in the Indian market. This huge demand-supply gap provides enough opportunity for

Market drivers

- ▶ Automotive electronics
- ▶ Consumer electronics
- ▶ Industrial electronics
- ▶ Light emitting diodes (LEDs)
- ▶ Strategic electronics
- ▶ Telecommunications

new manufacturing plants to come up in India.

Areas of concern

Doshi expresses his concerns about the time lag between announcement and implementation of government initiatives due to slow translation. It is a big challenge for achieving expected growth through those initiatives-related demand that would drive consumption and investment.

Muniswamy points out the main hindrances for starting PCB manufacturing in the country. These are high capital cost, non-availability of a good percentage of raw materials and world-class manufacturing equipment.

He also mentions that, typically in countries like Taiwan, China, Korea and Japan, PCB companies are located close to EMS or OEM companies and form a good ecosystem. In India, manufacturing itself is very small and is spread across the country from